Chapter 2: Back-of-the-Envelope Estimation

Introduction

Back-of-the-envelope estimation is a crucial skill in system design interviews. It involves making quick, rough calculations to assess system capacity or performance. According to Jeff Dean, Google Senior Fellow, these estimates help evaluate whether designs meet requirements through thought experiments and common performance benchmarks.

This chapter covers key concepts, methodologies, and examples to build proficiency in scalability and estimation.

Section 1: Key Concepts

Power of Two

Understanding data volume in terms of powers of two is fundamental:

Power	Approximate value	Full name	Short name
10	1 Thousand	1 Kilobyte	1 KB
20	1 Million	1 Megabyte	1 MB
30	1 Billion	1 Gigabyte	1 GB
40	1 Trillion	1 Terabyte	1 TB
50	1 Quadrillion	1 Petabyte	1 PB

This knowledge helps in performing accurate storage and bandwidth calculations.

Latency Numbers Every Programmer Should Know

Latency numbers represent the time taken for various operations in computing systems. These provide insights into relative performance:

Operation	Latency (2020)
L1 Cache Access	0.5 ns
L2 Cache Access	7 ns
Main Memory Access	100 ns
SSD Random Read	150 µs
HDD Random Seek	10 ms
Round-Trip in Data Center	500 µs
Inter-Region Data Center	150 ms

Key Insights:

- Memory is fast, disk is slow.
- Avoid disk seeks whenever possible.
- Compress data before transmitting over the internet to save bandwidth.

Availability Numbers

High availability (HA) ensures minimal downtime. Availability is expressed in **nines**:

- **99% (Two Nines):** ~3.65 days/year of downtime
- **99.9% (Three Nines):** ~8.8 hours/year of downtime
- **99.99% (Four Nines):** ~52 minutes/year of downtime
- **99.999% (Five Nines):** ~5.3 minutes/year of downtime
- **99.9999% (Six Nines):** ~31.56 seconds/year of downtime

Cloud providers like Amazon, Google, and Microsoft aim for SLAs (Service Level Agreements) of **99.9% or higher**.

Section 2: Example Estimation - Twitter QPS and Storage Requirements

Assumptions

- **300 million monthly active users (MAU).**
- **50% daily active users (DAU).**
- **Average tweets/user/day:** 2.
- **10% of tweets contain media.**
- **Data retention:** 5 years.

Estimations

1. **Query Per Second (QPS):**
    - DAU = (300M x 50% = 150M)
    - Tweets QPS = (150M x 2 tweets / 24 hour / 3600 seconds = ~3500)
    - Peak QPS = (2 x 3500 = ~7000)

2. **Media Storage:**
    - **Tweet Size Components:**
        - `tweet_id` : 64 bytes
        - `text` : 140 bytes
        - `media` : 1 MB
    - **Daily Media Storage:** (150M x 2 x 10% x 1MB = 30TB per day)
    - **5-Year Storage:** (30TB x 365 x 5 = ~55PB)

Section 3: Tips for Effective Estimation

1. Rounding and Approximation

Precision is not critical; focus on the process. Simplify complex calculations using round numbers. For example:

- (99987 / 9.1) can be approximated as (100,000 / 10 = 10,000).

2. Write Down Assumptions

Document assumptions clearly for future reference.

3. Label Units

Avoid ambiguity by labeling units (e.g., `5 MB` instead of `5`).

4. Common Estimation Scenarios

- **QPS (Queries Per Second):** Measure traffic intensity.
- **Peak QPS:** Account for traffic spikes.
- **Storage Requirements:** Estimate total data needs.
- **Cache Requirements:** Evaluate memory requirements for caching.
- **Number of Servers:** Calculate hardware needs based on workload.